MIS 3013 — Introduction to Programming

Section 004 | Spring 2026

Course Information

Instructor:	Adam Ackerman
Office:	Adams Hall 3213
Office Hours:	By Appointment
Email:	adam@ou.edu (include "MIS-3013" in subject line)
Location:	MFPH 3065
Meeting Times:	Mon/Wed/Fri 10:00 AM–10:50 AM
Course Site:	https://canvas.ou.edu
Final Exam:	Tue, May 12, 2026, 8:00 AM–10:00 AM, MFPH 3065

Prerequisites

MIS 2113 — Computer-Based Information Systems

It is expected that students have completed MIS 2113. We expect students to have a basic level of skill and experience with the Windows OS. We will not be spending class time reviewing basic use of these technologies.

Textbooks & Software

[Optional] *Textbook:* Beginning C# 7 Programming with Visual Studio 2017, 1st Edition

[Optional] *Textbook:* C# 10 and .NET 6 — Modern Cross-Platform Development (or C# 11 and .NET 7)

[Required] *Software:* Microsoft Visual Studio (Windows). Mac users must use Parallels and/or Boot Camp — a Windows environment is required for all programming work. — https://visualstudio.microsoft.com/

[Required] *Software:* GitHub account — https://github.com/

Course Description

This course introduces students to the fundamentals of computer programming with an emphasis on problem solving, logical thinking, and program design. Students learn how to analyze written problem descriptions and translate them into working programs using a structured development process. Core topics include programming syntax, control structures, methods, collections, and introductory object-oriented concepts. Through hands-on assignments, students gain experience writing, testing, and debugging programs using real-world integrated development environments and professional development tools, including cloud-based repositories for managing and submitting code. By the end of the course, students will be able to design and implement basic software solutions that meet given requirements and follow accepted programming practices.

Course Design

This class combines a variety of techniques to capture student interests from several possible angles. The readings and lectures provide a base of knowledge that students can use in other components of the course, future course work, and future career. In-class exercises provide an avenue for applying the knowledge base. Guest lectures provide further opportunities for discovery of how the material covered in class applies in everyday business life. This approach only works when students complete the readings prior to class and then participate in the exercises and discussions in class.

Course Modules

Module 1: Programming Fundamentals — *Covers variables, assignments and expressions, conditionals, loops, string manipulation, and collections. Students write, test, and debug programs using Visual Studio and GitHub.*

Module 2: Functions & Classes — *Introduces functions and object-oriented programming with classes. Students apply these concepts to design and implement structured applications.*

Module 3: File I/O, JSON & APIs — *Explores file input/output, working with JSON data, and consuming APIs. Students apply professional development practices culminating in the Module 3 / final exam.*

Course Level Outcomes (CLOs)

CLO-1: Apply fundamental programming syntax and constructs to analyze and solve basic business and computational problems. *[Assessed by: Homework 1, Homework 2, Homework 3, Module 1 Exam]*

CLO-2: Design and implement structured applications using core programming concepts including variables, control structures, methods, collections, and basic object-oriented principles. *[Assessed by: Homework 4, Homework 5, Homework 6, Module 2 Exam, Module 3 Exam]*

CLO-3: Translate written problem descriptions and business requirements into working programs by following a systematic problem-solving and program development process. *[Assessed by: Homework 1, Homework 2, Homework 3, Homework 4, Homework 5, Homework 6, Homework 7, Homework 8, Module 1 Exam, Module 2 Exam, Module 3 Exam]*

CLO-4: Use an integrated development environment to write, test, debug, and document programs effectively. *[Assessed by: Class Participation, Homework Assignments, Module 1 Exam, Module 2 Exam, Module 3 Exam]*

CLO-5: Utilize cloud-based version control tools (e.g., GitHub) to manage source code, submit programming assignments, and demonstrate professional development practices. *[Assessed by: Homework Assignments, Module 3 Exam]*

Grading & Assessments

Assessment Breakdown

Class Participation — 20% (In-class problems used to gauge learning)

Homework Assignments — 20% (Multiple assignments (see calendar); submit to the appropriate GitHub repository)

Module 1 Exam — 20%

Module 2 Exam — 20%

Module 3 Exam — 20% (Held during the final exam period.)

Class Participation

For the design of this class to work, students must be actively involved in the class and attend lectures. To ensure that active learning is happening, short problems will be provided during class and will be used to gauge learning and make sure that teaching is effectively being received and understood by the class. The answers to the problems will be evaluated in class and any concepts that are not understood will be explained.

Homework Assignments

Homework assignments for this course are designed to encourage students to work actively with the course material and thereby master course materials. Further, homework enables students and the professor to recognize any points that are not yet fully understood. Students are expected to complete each of the homework problems. If, after considerable effort, you are stuck on a problem, note (on the homework) the specific difficulty you are having and ask one of the TAs for help.

Students may discuss the homework problems and approaches to answering them with their classmates, TAs, and the professor. More importantly, it is essential that you understand the homework assignments to do well on the exams.

All assignments must be zipped and submitted to the appropriate GitHub repository.

Module 1 Exam

All exams will consist of programming problems like those assigned for homework, participation, as well as in-class problems. These exams are designed to assess your basic understanding of the concepts and their application as covered in class.

Module 2 Exam

All exams will consist of programming problems like those assigned for homework, participation, as well as in-class problems. These exams are designed to assess your basic understanding of the concepts and their application as covered in class.

Module 3 Exam

All exams will consist of programming problems like those assigned for homework, participation, as well as in-class problems. These exams are designed to assess your basic understanding of the concepts and their application as covered in class.

Grading Scale

- **A**: 90–100%
- **B**: 80–89%
- **C**: 70–79%
- **D**: 60–69%
- **F**: 0–59%

Course Policies

> ### Academic Integrity
>
> The University of Oklahoma has an Academic Misconduct Code that governs student academic performance in and out of the classroom. The Academic Misconduct Code can be reviewed online at http://www.ou.edu/integrity
>
> Academic misconduct is defined as "any act that improperly affects the evaluation of a student's academic performance or achievement". All work assigned in this course is to be completed individually (unless otherwise stated). The steps and procedures as outlined in the Academic

Misconduct Code will be followed in all cases of academic misconduct in this class.

MIS Division Guidelines for Academic Misconduct:

- All academic misconduct will be reported at: http://www.ou.edu/content/dam/integrity/docs/academic-misconduct-report.pdf
- The instructor will decide whether the misconduct is an Admonition or Violation of the Academic Integrity Code.
- When misconduct occurs on labs/similar assignments and is categorized as Admonition: a grade of zero will be recorded and a deduction equal to the value of the assignment will be applied.
- When misconduct occurs on an exam and is categorized as Admonition: a grade of zero will be recorded.
- When misconduct occurs on an exam and is categorized as a Violation: no grade penalty is imposed until the student is officially found responsible. Until then, "N" is filed as the final course grade. If established, a violation results in a grade penalty and a University penalty.
- Faculty may impose greater grade penalties than those recommended above.

Laptop Requirements

Students must bring a laptop to every class. iPads, Chromebooks, and tablets are not compatible with the required course software and cannot be used for assignments.

Mac Users: This course requires a Windows environment for all programming work. Mac users must install Parallels and/or Boot Camp. MacBooks older than 2 years often have issues with required software and grading tools. If you experience problems, you must rent a laptop from an OU IT kiosk to complete your work.

It is your responsibility to ensure your device is fully functional and compatible with course requirements.

All students majoring or minoring in MIS are expected to have a laptop computer that meets the minimum hardware and software criteria established by the MIS faculty. Recommended minimum laptop specifications for 2025–2026:

Hardware	Software
Processor: Intel Core i5 (12th gen or newer) / AMD Ryzen 5 (5000 series or newer) / Apple M1 or M2 equivalent	Operating System: Windows 11 (or higher) / macOS Ventura (or higher)
Memory (RAM): Minimum 16 GB	Productivity Suite: Microsoft Office 365 (student license provided by OU)
Storage: 512 GB SSD or higher	Database Tools: SQL Server Management Studio (SSMS) for Windows; Azure Data Studio for macOS/Linux
Wireless: Wi-Fi 6 (802.11ax) recommended; Wi-Fi 5 (802.11ac) acceptable	Browsers: Latest versions of Chrome
Ports: At least 2 USB ports	Antivirus (Windows Defender/McAfee free version)
Screen Size: 13" or larger recommended (1080p minimum)	
Battery Life: At least 6 hours on a single charge recommended	

We recommend that students purchase a PC/Windows laptop. Price College and the Library also have Dell laptops available to borrow from kiosks for class time and homework assignments.

Excused Absence Policy

Regular attendance is required. Absences will be considered excused only under the circumstances outlined below and only when appropriate documentation is submitted in a timely manner. The instructor retains sole discretion to determine whether an absence and its documentation meet the standards of this policy.

1. University-Recognized Excused Absences

A. Religious Observances

In accordance with University policy, absences for the observance of major religious holidays will be excused. Students must provide written notice as early as possible and no later than the end of the second week of the semester, or as soon as the conflict becomes known.

B. Official University-Sponsored Activities

Absences resulting from participation in official University-sponsored activities are excused. Written verification from the sponsoring University office is required. Advance notice is expected whenever practicable.

2. Medical Emergency

A medical emergency is a serious, acute condition requiring urgent or emergent medical treatment that renders the student unable to participate in academic activities.

Examples include, but are not limited to:

- Emergency room treatment
- Hospitalization
- Urgent surgical procedures
- Serious injury requiring immediate medical care

Required Documentation

Official documentation from a licensed healthcare provider or medical facility must include:

- Date(s) of treatment
- A statement confirming the student was medically unable to attend class on specified date(s)
- Provider signature or institutional verification

Documentation that merely confirms an appointment occurred, without stating incapacity to attend class, is insufficient.

Routine illness (e.g., cold, flu, minor viral illness, headache, fatigue, or similar conditions) does not qualify as a medical emergency and is not an excused absence under this policy, even with a doctor's note.

3. Motor Vehicle Accident

An absence resulting from a motor vehicle accident may be excused if the incident directly prevents class attendance.

Required Documentation

Students must provide a copy of the official police report documenting the accident. The report must clearly indicate the date and time of the incident and demonstrate that it coincided with the scheduled class time or otherwise made attendance not reasonably possible.

If medical treatment resulted from the accident, documentation meeting the requirements of the

Medical Emergency section may also be required.

4. Family Emergency

A family emergency is a significant, unforeseen event involving an immediate family member that requires the student's presence.

Immediate family member is defined as: spouse or domestic partner; child; parent or legal guardian; sibling; grandparent; grandchild; or corresponding in-law relationship.

Qualifying circumstances may include:

- Death of an immediate family member
- Serious medical emergency involving an immediate family member

Required Documentation

Appropriate third-party verification must be provided (e.g., obituary, funeral documentation, hospital verification, or written confirmation from a responsible official). Verification through the Dean of Students Office may be required.

5. Court Appearance

An absence may be excused when a student is legally required to appear in court and the appearance directly conflicts with scheduled class time.

Excused court appearances may include, but are not limited to:

- Jury duty
- Subpoenaed testimony
- Required court hearings

Voluntary legal meetings, consultations with attorneys, or court appearances scheduled at the student's discretion do not qualify.

Required Documentation

Students must provide official court documentation that clearly indicates:

- The date and time of the required appearance
- Verification that the student's presence was mandatory

Advance notice is expected whenever practicable.

6. Non-Qualifying Absences

The following do not constitute excused absences:

- Routine/chronic illness
- Personal travel
- Employment conflicts
- Oversleeping
- Minor appointments
- Technical difficulties
- Failure to plan appropriately

7. Notification and Timing

Students must notify the instructor as soon as reasonably possible. Required documentation must be submitted within one week of returning to class unless otherwise approved in writing. Failure to provide timely notice or adequate documentation may result in the absence being recorded as unexcused.

Getting Help from TAs

Students are strongly advised to get help from the TAs when having problems with the class material. TAs are available during class and have office hours (see Canvas for details). Their job is to help you when you get stuck, so feel free to approach them. They took this class recently and understand the problems you may run into well.

Timeliness

To receive full credit for a written assignment, the assignment must be turned in before the due date of the assignment. Late work will not be accepted. Participation scores will be given only to those who come to class. If you have obligations that conflict with exam or assignment due dates, you should make arrangements with the instructor as soon as possible to complete these ahead of time, but no later than 48 hours before the due date. Feel free to submit homework assignments early.

Grade / Score Appeals

Students wishing to appeal a grade or score must submit the appeal to the instructor **in writing within 48 hours** of receiving the graded work. Score changes are at the discretion of the instructor, not the TAs.

It is important to understand that your score may go up or down based on a complete review of the work in question. It is usually the case that changing a few points on an assignment rarely makes a difference in the final grade. As such, student (and instructor) time is generally much better spent discussing and clarifying the information content presented in the course.

Missed Exams / Make-Up Policy

Make-up exams will be given only in extraordinary circumstances. If you expect to miss an exam or are unable to meet another requirement, please discuss this with the instructor **before the scheduled date, no later than 48 hours before the due date**.

University Policies

Mental Health Support Services

As a university student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. OU offers services to assist you with addressing these and other concerns you may be experiencing. You can learn more about the broad range of confidential mental health services available on campus via the Counseling, Wellness, and Prevention Services (CWPS) at https://www.ou.edu/health/cwps, (405) 325-2911 or at Goddard Health Center, 620 Elm Avenue. Additionally, you can also access TimelyCare, a virtual health and well-being platform that provides 24/7/365 access to medical care, mental health support, and wellness services. To connect with TimelyCare, visit timelycare.com/OU or download the app.

Title IX / Sexual Misconduct

The University of Oklahoma is committed to providing a safe learning environment for all members of our community. Title IX of the Education Amendments of 1972 prohibits sex discrimination in educational programs and activities. This includes sexual harassment, sexual assault, domestic violence, dating violence, and stalking.

As your instructor, I am a **mandatory reporter**. This means that if you share information about sexual misconduct or harassment with me — whether in person, in writing, in class discussions, on Canvas, or in any other setting — I am required to report that information to the OU Institutional Equity Office. If you need confidential support, you may contact **OU Advocates** at (405) 615-0013. You can also reach the Institutional Equity Office at (405) 325-3546 or visit https://www.ou.edu/eoo.

Reasonable Accommodation

The University of Oklahoma is committed to making its activities accessible to all participants. If you have a disability and require accommodations in this course, please contact the **Accessibility and Disability Resource Center (ADRC)** through iAdvise to schedule an appointment at https://www.ou.edu/adrc, by emailing adrc@ou.edu, or by calling (405) 325-3852. Accommodations are not retroactive. You must provide your accommodation letter to me as early as possible in the semester to ensure appropriate arrangements can be made.

Religious Observance

It is the policy of the University to excuse the absences of students that result from religious observances and to provide without penalty for the rescheduling of examinations and additional required classwork that may fall on religious holidays [Faculty Handbook 3.15.2]. If a religious holiday conflicts with a class meeting, quiz, exam, or assignment deadline, please notify me at least one week in advance so that we can make appropriate arrangements.

Pregnancy Adjustments

Title IX protects students from discrimination based on pregnancy, childbirth, or related conditions. Students who are pregnant or experiencing pregnancy-related conditions may request reasonable adjustments. For assistance, contact the **Institutional Equity Office** at (405) 325-3546 or visit https://www.ou.edu/eoo.

Final Exam Preparation Period

The 7 calendar days before the first day of the final exam period are designated as the final exam preparation period. During this time, no major homework, projects, presentations, or other graded work may be assigned that is due before the final exam period begins. Instructors may continue to cover new material during this period. Please refer to the Academic Calendar for the specific dates each semester.

Emergency Protocol — Severe Weather

In the event of severe weather, follow OU Alert instructions. If a tornado warning is issued while you are on campus:

- **Get In** — Move to the nearest interior shelter location.
- **Get Down** — Go to the lowest floor possible.
- **Cover Up** — Protect your head and neck.

Building-specific severe weather refuge maps are posted throughout campus. Sign up for **OU Alert** at https://www.ou.edu/oupd/oualert to receive emergency notifications.

Emergency Protocol — Active Threat (Run, Hide, Fight)

In the event of an active threat on campus:

- **Run** — If there is a safe escape path, leave immediately and call 911.
- **Hide** — If evacuation is not possible, find a secure location, lock/barricade the door, and silence your phone.
- **Fight** — As a last resort, use whatever means necessary to disrupt and disarm the threat.

OU Alerts are sent via the one.ou.edu system. Emergency contacts: OUPD Norman (405) 325-1717 / (405) 325-1911; OUPD Tulsa (918) 660-3900 / (918) 660-3333.

Emergency Protocol — Fire Alarm

In the event a fire alarm sounds:

1. **Leave** the building immediately.
2. **Know** your nearest exit route.
3. **Assist** those who need help evacuating, if it is safe to do so.

4. **Proceed** to the designated assembly area.
5. **Notify** emergency personnel of anyone who could not evacuate.
6. **Wait** for official clearance before re-entering the building.

Why You Belong at the University of Oklahoma

The University of Oklahoma is committed to creating a diverse and inclusive community where all students feel welcomed, valued, and supported. We recognize and respect the unique backgrounds, perspectives, and experiences that each student brings to our campus. Regardless of your race, ethnicity, gender identity, sexual orientation, religion, socioeconomic status, age, disability, or any other aspect of your identity, you belong here.

We strive to foster an environment that celebrates diversity and promotes understanding, empathy, and respect. We encourage you to embrace your own identity and to appreciate the richness that comes from engaging with others who are different from you. By learning from one another, we can collectively grow and create a more equitable and just world.

If you ever feel that you don't belong or experience any form of discrimination or harassment, please know that there are resources available to support you. You are not alone, and we are here to help you navigate any challenges you may encounter. Together, let's build a community where everyone can thrive and succeed.

Course Reflection Survey

Near the end of the semester, the university will administer an online course reflection survey. Your feedback helps improve instruction and course design. Surveys are confidential; individual responses are not released to instructors until after final grades have been submitted. I encourage you to take a few minutes to share your honest perspective.

Copyright Statement

Course materials prepared by the instructor are the intellectual property of the instructor. Students are expressly prohibited from selling notes or other course materials to any person or commercial firm without the express written permission of the instructor. Recording of lectures or class sessions is permitted only with the instructor's permission. Any approved recordings are

> for personal academic use only and may not be distributed, published, or posted publicly in any form.

Generative AI Policy

> I permit the use of standalone Generative AI tools like ChatGPT, Copilot, Claude, etc., for the following:
>
> - Searching for publications on your research topic
> - Creating an outline, text, or speech for presentations and writeups
> - Providing grammatical feedback on drafts
> - Further study on concepts outside of class
> - Troubleshooting problems with software
> - Creating initial ideas for projects
>
> Whenever you use Generative AI for any of these, you are required to make a declaration acknowledging how and to what extent you used it. Please include an appendix that details the prompts you used and the responses you obtained. Usage of Generative AI outside of the clearly defined scope, and without acknowledgment as required, will be considered a violation of the academic integrity policy for this course.
>
> **Note:** Regardless of the above, no AI tools of any kind are permitted during quizzes, exams, or any other timed in-class assessments unless the instructor explicitly states otherwise for a specific assessment.

Course Calendar

The schedule below is tentative; adjustments may be made. Announcements in class, via email, or on Canvas supersede this outline.

Wk	Mon	Wed	Fri
1	Mon, Jan 12, 2026 **Introduction, Environment Setup** *Visual Studio, GitHub*	Wed, Jan 14, 2026 **Variables, Assignments & Expressions**	Fri, Jan 16, 2026 **Conditionals** *Lecture, In-class Example*
2	Mon, Jan 19, 2026	Wed, Jan 21, 2026	Fri, Jan 23, 2026

Wk	Mon	Wed	Fri
	Martin Luther King Jr. Day *No Class*	**Conditionals** *Participation*	**Conditionals** *Participation*
3	Mon, Jan 26, 2026 **Loops** *Lecture, In-class Example*	Wed, Jan 28, 2026 **Loops** *Participation*	Fri, Jan 30, 2026 **Loops** *Participation* Homework 1
4	Mon, Feb 2, 2026 **Loops** *Lecture, In-class Example*	Wed, Feb 4, 2026 **Loops** *Participation*	Fri, Feb 6, 2026 **Loops** *Participation*
5	Mon, Feb 9, 2026 **String Manipulation** *Lecture, In-class Example*	Wed, Feb 11, 2026 **String Manipulation** *Participation*	Fri, Feb 13, 2026 **String Manipulation** *Participation* Homework 2
6	Mon, Feb 16, 2026 **Collections** *Lecture, In-class Example*	Wed, Feb 18, 2026 **Collections** *Participation*	Fri, Feb 20, 2026 **Collections** *Participation* Homework 3
7	Mon, Feb 23, 2026 **Intro to Module II** *Setup accounts + Participation over Module 1*	Wed, Feb 25, 2026 **Module 1 Review** *Variables, String, Loops, Collections* Homework 4	Fri, Feb 27, 2026 **Module 1 Exam**
8	Mon, Mar 2, 2026 **Functions** *Lecture, In-class Example*	Wed, Mar 4, 2026 **Functions** *Participation*	Fri, Mar 6, 2026 **Functions** *Participation*
9	Mon, Mar 9, 2026 **Classes** *Lecture, In-class Example*	Wed, Mar 11, 2026 **Classes** *Participation*	Fri, Mar 13, 2026 **Classes** *Participation* Homework 5
10	Mon, Mar 16, 2026 **Spring Break** *No Class*	Wed, Mar 18, 2026 **Spring Break** *No Class*	Fri, Mar 20, 2026 **Spring Break** *No Class*
11	Mon, Mar 23, 2026 **Classes** *Participation*	Wed, Mar 25, 2026 **Classes** *Participation*	Fri, Mar 27, 2026 **Classes** *Participation*
12	Mon, Mar 30, 2026 **Classes** *Participation*	Wed, Apr 1, 2026 **Module 2 Review** *Functions, Classes* Homework 6	Fri, Apr 3, 2026 **Module 2 Exam**
13	Mon, Apr 6, 2026 **File I/O** *Lecture, In-class Example*	Wed, Apr 8, 2026 **File I/O** *Participation*	Fri, Apr 10, 2026 **File I/O** *Participation*

Wk	Mon	Wed	Fri
14	Mon, Apr 13, 2026 **JSON** *Lecture, In-class Example*	Wed, Apr 15, 2026 **JSON** *Participation*	Fri, Apr 17, 2026 **JSON** *Participation* Homework 7
15	Mon, Apr 20, 2026 **API** *Lecture, In-class Example*	Wed, Apr 22, 2026 **API** *Participation*	Fri, Apr 24, 2026 **API** *Participation*
16	Mon, Apr 27, 2026 **API** *Participation*	Wed, Apr 29, 2026 **API** *Participation*	Fri, May 1, 2026 **Module 3 Review** *File I/O, API* Homework 8 & 9
17	Mon, May 4, 2026	Wed, May 6, 2026	Fri, May 8, 2026
Final	**FINAL EXAM** Tue, May 12, 2026 · 8:00 AM – 10:00 AM · Room: MFPH 3065		